Summary in Graph

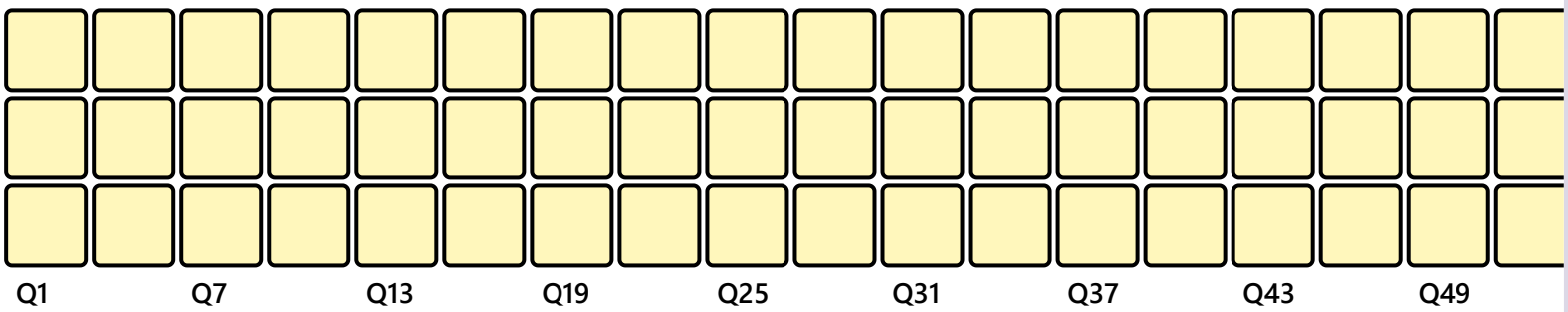
Exam Summary (GO Classes CS Test Series 2026 | Mixed Subjects | Full Length Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (25) out of 252 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Probability

Let $Z = 3X - Y - 5$, where X and Y are independent random variables with $\mathrm{Var}(X) = 1$ and $\mathrm{Var}(Y) = 2$. What is $\mathrm{Var}(Z)$?

A. 4
B. 7
C. 11

D. None of the above.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose a digital communications channel has a probability of bit error of p, where all errors are independent of each other. If you transmit N bits, what's the probability that exactly k bits are received incorrectly?

A. $\binom{N}{N-k} p^{N-k}(1-p)^k$
B. $p^{k-1}(1-p)$
C. $(1-p)^{k-1}p$
D. $\binom{N}{k} p^k(1-p)^{N-k}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

You ask your neighbor to water a sickly plant while you are on vacation. Without water, the plant will die with a probability 0.8. With water, it will die with a probability 0.15. You are 90 percent certain the neighbor will remember to water the plant.
What is the probability that the plant will be alive when you return?

A. 0.85
B. 0.79
C. 0.90
D. 0.63

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Given two Boolean random variables, A and B, where $P(A) = 1/2, \quad P(B) = 1/3$, and $\mathrm{P}(A \mid \neg B) = 1/4$, what is $\mathrm{P}(A \mid B)$?

A. $1/6$
B. $1/4$
C. $3/4$
D. 1

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Let A, B denote events in a sample space S and let $\mathrm{P}(.)$ denote a probability measure on S. Which of the following is/are TRUE?

A. If $P(A \mid B) = P(A)$ then $P(B \mid A) = P(B)$.
B. If $P(A) \leq P(B)$ then $A \subset B$.

C. If $A \subset B$ then $P(A) \leq P(B)$
D. $P(A \mid B') = 1 - P(A \mid B)$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Numerical Type | Award: 1 | Penalty: 0 | Probability

Consider now a refined probabilistic model with $n = 2$ questions. For $i \in \{1, 2\}$, let X_i be an indicator random variable which is 1 iff question i is answered correctly by the student. We know $\mathbf{P}[X_1 = 1 \mid X_2 = 0] = 0.4, \mathbf{P}[X_2 = 1 \mid X_1 = 1] = 0.8$ and $\mathbf{P}[X_1 = 1] = 0.5$ Compute the $P[X_2 = 1 \mid X_1 = 0]$

Your Answer: | Correct Answer: 0.7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $\mathrm{AX} = 0$ be a system of linear equations, where $\mathrm{A} = [a_{ij}]_{m \times n}$, X is a column matrix and 0 is a null matrix. If $\mathrm{Rank}(A) = r$

A. Exactly r linearly independent solutions
B. Exactly n linearly independent solutions
C. Exactly $n - r$ linearly independent solutions
D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -7 \\ 4 \\ 2 \end{bmatrix}$. Which of the following statements must be true?

I. $A = \begin{bmatrix} -7 & 0 & 0 \\ 4 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$ is of rank 1 such that $A\mathbf{x} = \mathbf{v}$.

II. $A = \begin{bmatrix} -7 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & -1 & 0 \end{bmatrix}$ is of rank 2 such that $A\mathbf{x} = \mathbf{v}$.

III. $A = \begin{bmatrix} -7 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2/5 \end{bmatrix}$ is of rank 3 such that $A\mathbf{x} = \mathbf{v}$.

A. Only I
B. Only II
C. I, II
D. All of them

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose that A is $n \times m$. Which of the following statements must be true?

I. If $\text{Rank}(A) = n$ and $m > n$ (i.e., $m \geq n+1$), then the system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
II. If $\text{Rank}(A) = m$ and $m \leq n$, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.
III. If $\text{Rank}(A) = m$ and $m \leq n$, then $A\mathbf{x} = \mathbf{b}$ has at most one solution.
IV. If $\text{Rank}(A) = n$ and $m > n$, (i.e., $m \geq n+1$), then $A\mathbf{x} = \mathbf{b}$ has infinitely many solution.

A. I, II, III
B. II, IV
C. I, III, IV
D. I, II, IV

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Find an equation relating a, b and c so that the linear system

$$\begin{aligned} x + 2y - 3z &= a \\ x - y + 8z &= b \\ -6x - 3y - 15z &= c \end{aligned}$$

is consistent for any values of a, b, c satisfying that equation.

A. $3a - 3b + c = 0$
B. $3a + 3b + c = 0$
C. $3a + 3b - c = 0$
D. $6a - 3b - c = 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

For what value of c are the vectors $\begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ c \end{pmatrix}$ in $\mathbb{R}^3$ linearly dependent?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the following augmented matrix where $*$ can be any value that is not 0 and not 1 :

$$\left[\begin{array}{cccc|c} 1 & * & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & 1 & * & * \end{array}\right]$$

The system of linear equations represented:

A. has a unique solution
B. has no unique solution
C. has infinite solutions
D. has no solutions

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

A matrix A can be row reduced to the following echelon form

$$\begin{bmatrix} 1 & 1 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & a & b \\ 0 & 0 & 0 & 0 & c \end{bmatrix}$$

Depending on the values of the constants a, b and c the rank of A

A. must be 2
B. must be 3
C. must be 4
D. can be $2, 3$ or 4

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

For a connected, undirected graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$, which of the following must be true?

I. $\sum_{v \in \mathrm{V}}$ degree (v) is even.
II. $|\mathrm{E}| \geq |\mathrm{V}| - 1$
III. G has at least one vertex with degree 1.

A. I only
B. I, II only
C. I, III only
D. I, II, III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

An Euler circuit of an undirected graph is a circuit in which each edge of the graph appears exactly once. Which of the following undirected graphs must have an Euler circuit?

I. A complete graph with 12 vertices
II. A complete graph with 13 vertices
III. A tree with 13 vertices

A. I only
B. II only
C. III only
D. I and II Only

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

A full binary tree is a rooted tree in which every internal node has exactly two children. How many internal nodes are there in a full binary tree with 500 leaves?

A. 250
B. 499
C. 500
D. 501

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Let G be a non-cyclic group of order 57. Then the number of elements of order 5 in G is _____

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

The number of different Total order relations on a set of 7 elements is _____

Your Answer: | Correct Answer: 5040 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

Consider a 5×6 rectangular grid as depicted in Figure 1. A staircase walk is a path in the grid which moves only right or up.

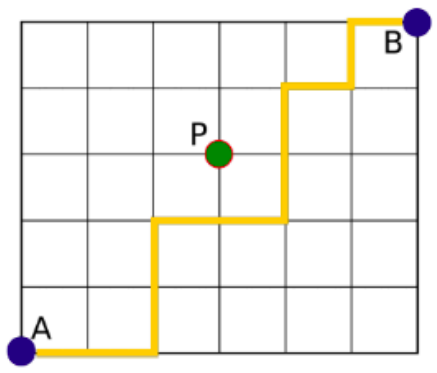


Figure 1: The 5×6 grid and a valid staircase walk from A to B avoiding P.

What is the total number of staircase walks from A to B that may or may not pass through the point P?

Your Answer: | Correct Answer: 462 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Consider the statement form S :

$$p \vee (\sim p \wedge q) \Rightarrow \sim q.$$

Which of the following is true?

A. S is a tautology
B. S is a Contradiction
C. S is a Contingency
D. Antecedent of S is equivalent to consequent of S

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

What is the order of smallest non-cyclic group?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let $\mathcal{P}(A)$ denote the power set of A. If $\mathcal{P}(\mathrm{A}) \subseteq \mathrm{B}$ then

A. $2^{|\mathrm{A}|} \leq |\mathrm{B}|$
B. $2^{|\mathrm{A}|} \geq |\mathrm{B}|$
C. $2|\mathrm{A}| < |\mathrm{B}|$
D. $|\mathrm{A}| + 2 \leq |\mathrm{B}|$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 1 | Penalty: 0 | Combinatory

The number of partitions of the set $\mathrm{X} = \{a, b, c, d\}$ with "a" and "b" in the same block is _____

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

Given a graph G is a connected planar simple graph with ' e ' edges and ' v ' vertices, with $v \geq 3$ and if all cycles in G have length ≥ 4, then which of the following relations hold always?

A. $e \leq 2v - 4$
B. $e \leq 2v - 2$
C. $e \leq 3v - 4$
D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Which of the following statements is TRUE?

A. For all sets A, B, and C, $A - (B - C) = (A - B) - C$.
B. For all sets A, B, and C, $(A - B) \cap (C - B) = (A \cap C) - B$.
C. For all sets A, B, and C, $(A - B) \cap (C - B) = A - (B \cup C)$.
D. For all sets A, B, and C, if $A \cap C = B \cap C$ then $A = B$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

To show that the circuit corresponding to the Boolean expression $(P \wedge Q) \vee (\sim P \wedge Q) \vee (\sim P \wedge \sim Q)$ can be represented using two logical gates, one shows that this Boolean expression is equal to $\sim P \vee Q$. The circuit corresponding to $(P \wedge Q \wedge R) \vee (\sim P \wedge Q \wedge R) \vee (\sim P \wedge (\sim Q \vee \sim R)$ computes the same function as the circuit corresponding to

A. $(P \wedge Q) \vee \sim R$
B. $P \vee (Q \wedge R)$
C. $\sim P \vee (Q \wedge R)$
D. $(P \wedge \sim Q) \vee R$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

The power set $\mathcal{P}((A \times B) \cup (B \times A))$ has the same number of elements as the power set $\mathcal{P}((A \times B) \cup (A \times B))$ if and only if

A. $A = B$
B. $A = \emptyset$ or $B = \emptyset$
C. $B = \emptyset$ or $A = B$
D. $A = \emptyset$ or $B = \emptyset$ or $A = B$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Number of Linearly independent eigenvectors corresponding to the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & -1 \\ 4 & 0 & 3 \end{pmatrix}$$

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Which of the following formulas is a formalization of the sentence: "Every connected and circuit-free graph is a tree". Where cir(x) means " x has circuit"

A. $\forall x[(\text{conn}(x) \wedge \neg \text{cir}(x)) \rightarrow \text{tree}(x)]$
B. $\neg \exists x[(\text{conn}(x) \wedge \neg \text{cir}(x)) \wedge \neg \text{tree}(x)]$
C. $\forall x[(\text{conn}(x) \wedge \neg \text{cir}(x)) \wedge \text{tree}(x)]$

D. $\forall x[(\text{conn}(x) \wedge \text{cir}(x)) \rightarrow \text{tree}(x)]$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let $[P(S), \supseteq]$ be a partial order relation defined on $P(S)$, where $S = \{1, 2, 3, 4\}$. $P(S)$ is the set of all subsets of S. Which of the following is the join of $\{1, 2\}, \{3, 4\}$?

A. $\{3, 4\}$
B. $\{1, 2, 3, 4\}$
C. $\varnothing$
D. $\{1, 2\}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the matrix $\mathbf{A} \in \mathbb{R}^{2\times 2}$, which takes a vector $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ and scales its y component by 2, but leaves the x component unchanged. In other words, $\mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 2y \end{bmatrix}$.

What are the eigenvalues of $\mathbf{A}$?

A. $1, 0$
B. $2, 0$
C. $1, 2$
D. $-1, 2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Suppose $A = [\,\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n\,]$ is an $n \times n$ invertible matrix, and $\mathbf{b}$ is a non-zero vector in $\mathbb{R}^n$. Which of the following statements is / are true?

A. $\mathbf{b}$ is a linear combination of $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$.
B. The determinant of A is nonzero.
C. $\text{rank}\, A = n$.
D. If $A\mathbf{b} = \lambda\mathbf{b}$ for some constant λ, then $\lambda \neq 0$.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the matrix $A = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 2 & 4 & -2 \end{bmatrix}$ whose row echelon form is $\begin{bmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{bmatrix}$. Find $a + b + c + d$:

A. -7

B. -4
C. -3
D. -1

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Suppose you wish to determine whether a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is linearly independent. You form the matrix $A = [\,\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{v}_4\,]$ and you calculate its reduced row echelon form, $R = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

You now decide to write $\mathbf{v}_2$ as a linear combination of $\mathbf{v}_1, \mathbf{v}_3$, and $\mathbf{v}_4$. Which is /are correct linear combination?

A. $\mathbf{v}_2 = \mathbf{v}_3 - 2\mathbf{v}_4$
B. $\mathbf{v}_2 = -3\mathbf{v}_3 - \mathbf{v}_4$
C. $\mathbf{v}_2 = \mathbf{v}_4 - 3\mathbf{v}_3$
D. $\mathbf{v}_2 = -\mathbf{v}_1 + \mathbf{v}_4$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A 5-bit codeword is sent over a noisy channel, where bits are flipped with probability p, each independently. An error-correcting code is designed such that if two or fewer bits of a codeword are flipped, the codeword can be correctly decode.

What's the probability a codeword cannot be decoded?

A. $\binom{5}{2} p^2(1-p)^3$
B. $p^5(1-p)$
C. $\binom{5}{3} p^3(1-p)^2 + 5p^4(1-p) + p^5$
D. $\binom{5}{3} p^3(1-p)^2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X be a random variable that takes values from 0 to 9 with equal probability $1/10$.

Let Z be a random variable $Z = (5 \bmod (X+1))$.

Find $P(Z=1)$?

A. $1/10$
B. $2/10$
C. $5/10$
D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let A and B are two events not necessarily independent.

$$P(A, B) = 0.15$$
$$P\left(A^c, B^c\right) = 0.45$$
$$P(A) = 0.6$$

What is the probability of $P\left(A^c, B\right)$?

Your Answer: | Correct Answer: 0.1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider the following scenario:
A box contains three coins: two regular coins and one fake two-headed coin, which always lands heads up.
If you randomly pick a coin from the box and toss it, what is the probability that it lands heads up?

A. $1/2$
B. $2/3$
C. $1/3$
D. $3/4$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Anne and Betty go fishing. Assume they catch fish independently and that each has a probability $0 < p < 1$ of catching no fish. Let A be the event that Anne catches no fish and B be the event that Betty catches no fish.

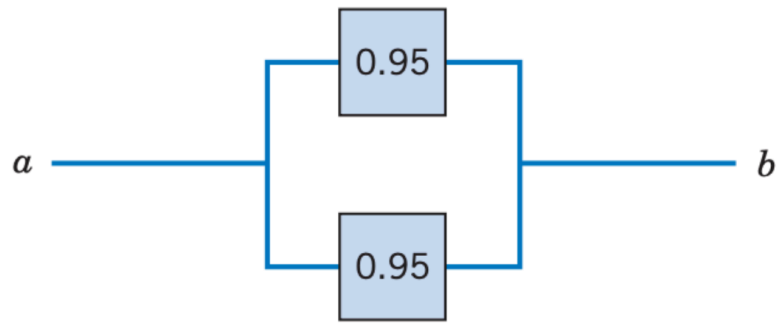
Find the conditional probability that Anne catches no fish given that at least one of them catches no fish i.e. $P(A \mid (A \cup B))$.

A. $1/(1-p)$
B. $1/(2-p)$
C. $1/p$
D. None of these

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

We have a circuit with two devices between points a and b. The circuit will work if there is at least one path between a and b. The probability of each device working is shown on the graph, and we assume independent device failures. What is the probability of the circuit functioning correctly?



A. 0.905
B. 0.998
C. 0.915
D. 0.940

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let G be a finite group. Let "1" denote the identity element of G. If g is an element of a finite group G, then the order of g is the smallest positive integer n such that $g^n = 1$, and it is denoted $o(g) = n$. The order of group G is denoted by $o(\mathrm{G})$ or $|\mathrm{G}|$.

Which of the following statements is necessarily true?

A. If $g^m = 1$, for some $g \in \mathrm{G}$, then $o(g) \mid m$.
B. If $g^m = 1$, for some $g \in \mathrm{G}$, then $o(g) = m$.
C. If $g^m = 1$, for some $\mathrm{g} \in \mathrm{G}$, then $m \mid o(\mathrm{g})$.
D. If $g^m = 1$, for some $g \in \mathrm{G}$, then $m \mid o(\mathrm{G})$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

If G is a group and $g \in \mathrm{G}$, then the subgroup generated by g is

$$\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$$

The set of all positive integers less than n and relatively prime to n with multiplication $\mod \mathrm{n}$ is a group denoted U_n.
i.e. $\mathrm{U}_n = \{g \in Z_n \mid \gcd(g, n) = 1\}$.

Which of the following is/are true about U_{20}?

A. In U_{20}, the size of the subgroup generated by 3 is 6.
B. U_{20} is an abelian group.
C. U_{20} is a cyclic group.
D. The order of an element y in a group G is equal to the size of the subgroup generated by y.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let G be a group and $g \in \mathrm{G}$. Let "1" denote the identity element of G. The order of $g \in \mathrm{G}$ is denoted $\mathrm{o}(\mathrm{g})$. The order of group G is denoted by $\mathrm{o}(\mathrm{G})$ or $|\mathrm{G}|$.

Which of the following statements is/are true?

A. $g^2 = 1$ if and only if $o(g) = 1$ or 2.
B. $o(g) = o\left(g^{-1}\right)$.
C. If $o(g) = n$ for some $n \in \mathrm{Z}^+$, then $g^{(-1)} = g^{(n-1)}$.
D. $o(g) = o\left(hgh^{-1}\right)$ for any $h \in \mathrm{G}$.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let G be a group. Let "1" denote the identity element of G. Which of the following is/are true?

A. if $a^{-1} = a$ for all $a \in \mathrm{G}$, then G is abelian.
B. if for all $a, b, c \in \mathrm{G}, ab = ca \Rightarrow b = c$, then G is abelian
C. If $(a * b)^{-1} = \left(b^{-1}\right) * \left(a^{-1}\right)$ for every $a, b \in \mathrm{G}$,,= then G is abelian.
D. $(ab)^n = a^n b^n$ for all $a, b \in \mathrm{G}$, $n \in Z$ if and only if G is abelian.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let $\mathrm{G} = \langle \{e, a, b, c, d\}, * \rangle$ be a group with the following Cayley table partially completed.

$*$	e	a	b	c	d
e	e	a	b	c	d
a	a		e	b	
b	b				
c	c				
d	d				

Which of the following is/are true for the given group G?

A. G is a cyclic group.
B. G is abelian.
C. The number of generators in G is 4.
D. The order of d is 2.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

A k-ary tree is a tree in which every node has at most k children. In a k-ary tree with n nodes and height h, which of the following is an upper bound for the maximum number of leaves as a function of h, k, and n? (height of the tree is the number of edges in the longest path from root to some leaf)

A. $\log_k n$
B. $n / \log_k(n)$
C. k^h
D. h^k

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

A graph is said to be 2-colourable if each vertex can be coloured either red or blue and no two vertices of the same colour are connected by an edge. If some graph is not 2-colorable, then we can reduce it to become 2-colorable by deleting some edges. We are given any simple graph with 101 nodes. k is the least required number of edges we have to delete in order to make this graph 2-colorable (Eg: $k = 0$ for a graph which is already 2-colorable). The minimum value for ' k ' to reach the worst case is _____

Your Answer: | Correct Answer: 2500 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which of the following statements is/are True?

A. $\{2,3,4\} \subseteq \mathrm{A}$ implies that $2 \in \mathrm{A}$ and $\{3,4\} \subseteq \mathrm{A}$.
B. $\{2,3,4\} \in \mathrm{A}$ and $\{2,3\} \in \mathrm{B}$ implies that $\{4\} \subseteq \mathrm{A}$ - B.
C. $\mathrm{A} \cap \mathrm{B} \supseteq \{2,3,4\}$ implies that $\{2,3,4\} \subseteq \mathrm{A}$ and $\{2,3,4\} \subseteq \mathrm{B}$.
D. A - $\mathrm{B} \supseteq \{3,4\}$ and $\{1,2\} \subseteq \mathrm{B}$ implies that $\{1,2,3,4\} \subseteq \mathrm{A} \cup \mathrm{B}$.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let A be any set.
The following are the definitions of Irreflexive and Asymmetric Relations :

a. A binary relation R on a set $\mathcal{A}$ is irreflexive iff $(x,x) \notin \mathrm{R}$ for all $x \in \mathcal{A}$.
b. A binary relation R on a set $\mathcal{A}$ is asymmetric iff whenever $(x,y) \in \mathrm{R}$, then $(y,x) \notin \mathrm{R}$.

We define a "strict order relation" as follows :
A binary relation R on a set A is a strict order on A iff R is irreflexive and transitive.
Which of the following is/are true?

A. Every strict order is asymmetric.
B. Every strict order is total order relation.
C. Every strict order is antisymmetric.
D. Every relation which is both Irreflexive and Asymmetric, is strict order.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Find the number of positive integers $n \in N$, such that $1 \leq n \leq 1000$, which are not divisible by either $4, 10$ or 15.

Your Answer: | Correct Answer: 667 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

Let $n \in \mathbb{N}$,

Consider the following two combinatorial identities :

1. $\binom{2n}{2} = 2\binom{n}{2} + n^2.$
2. $\binom{n}{0} + 2^1\binom{n}{1} + 2^2\binom{n}{2} + \ldots + 2^k\binom{n}{k} + \ldots + 2^n\binom{n}{n} = 3^n.$

Which of the above are correct?

A. Only 1
B. Only 2
C. Both
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Which of the following propositional logic formulas is/are valid?

A. $(p \to q) \wedge \neg q \to \neg p$
B. $(p \to q) \to (p \to \neg q)$
C. $(p \vee q \to \mathrm{r}) \vee p \vee q$
D. $(p \vee q) \wedge (p \to r \wedge q) \wedge (q \to \neg r \wedge p)$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let $f : \mathrm{X} \to \mathrm{Y}$ and $g : \mathrm{Y} \to \mathrm{Z}$. Let $h = g \circ f : \mathrm{X} \to \mathrm{Z}$. Suppose g is one-to-one and onto. Which of the following is FALSE?

A. If f is one-to-one then h is one-to-one and onto.
B. If f is not onto then h is not onto.
C. If f is not one-to-one then h is not one-to-one.
D. If f is one-to-one then h is one-to-one.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #54 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose $\mathrm{Z}_1, \mathrm{Z}_2, \mathrm{Z}_3$ are independent Bernoulli random variables with parameter $p = \frac{1}{2}$.

The random variable is defined as follows:

$$\mathrm{S} = \begin{cases} \mathrm{Z}_2\mathrm{Z}_3 & \text{if } \mathrm{Z}_1 = 1, \\ \mathrm{Z}_2 + \mathrm{Z}_3 & \text{if } \mathrm{Z}_1 = 0. \end{cases}$$

Find $\mathrm{P}(\mathrm{S} = 1)$.

A. $3/8$
B. $1/2$
C. $1/8$
D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose Y has a discrete distribution with pmf given by

y	0	1	2	3
$p_{\mathrm{Y}}(y)$	$1/2$	$1/4$	$1/8$	$1/8$

Calculate $\mathrm{E}\left[(\mathrm{Y}-1)^2\right]$?

A. $1/4$
B. $1/8$
C. $9/8$
D. $3/4$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #56 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A parallel system functions whenever at least one of its components works. Consider a parallel system of n components and suppose that each component works with probability p independently.
What is the probability the system is functioning?

A. $(1-p)^n$
B. p^n
C. $1-(1-p)^n$
D. $1-p^n$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #57 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be an $n \times n$ matrix. Its only eigenvalues are $1, 2, 3, 4, 5$, possibly with multiplicities.

What is the nullity of the matrix $\mathrm{A}+\mathrm{I}_n$, where I_n is the $n \times n$ identity matrix?

A. 0
B. 1
C. $n-1$
D. $n-5$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #58 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Which of the following statements is true for every matrix $\mathbf{M}$ with m rows and n columns, where $m < n$?

A. The system $\mathbf{Mx} = \mathbf{0}$ has no solution.
B. The system $\mathbf{Mx} = \mathbf{0}$ has infinitely many solutions.
C. For every $\mathbf{b}$ in $\mathbb{R}^m$, the system $\mathbf{Mx} = \mathbf{b}$ has infinitely many solutions.
D. For some $\mathbf{b}$ in $\mathbb{R}^m$, the system $\mathbf{Mx} = \mathbf{b}$ has a unique solution.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #59 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $\mathrm{A} = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$, and let B be a 2×3 matrix. Suppose that the second row of $(\mathrm{A}\ \mathrm{B})^{\mathrm{T}}$ is $\mathbf{v} = [-1 \quad 3]$ Compute the second column of B.

A. $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$
B. $\begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$
C. $\begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix}$
D. $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #60 | Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

The set $\{1, 2, 3, \ldots, n-1\}$ is a group under multiplication modulo n. Then the smallest value of n between 20 and 30 is $(20, 30$ included $)$ _____

Your Answer: | Correct Answer: 23 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #61 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which of the following functions are one-to-one but not onto.

A. $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = e^x$
B. $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(n) = n^2 + 3$
C. $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \sin x$
D. $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = x^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #62 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Assume the specifications

- $F(x, y)$: x is the father of y
- $M(x, y)$: x is the mother of y.

Which of the formulas in predicate logic below express the sentence 'Everybody has a mother'?

A. $\exists x \forall y M(x, y)$
B. $\exists y \forall x M(x, y)$
C. $\forall x \exists y M(x, y)$
D. $\forall y \exists x M(x, y)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

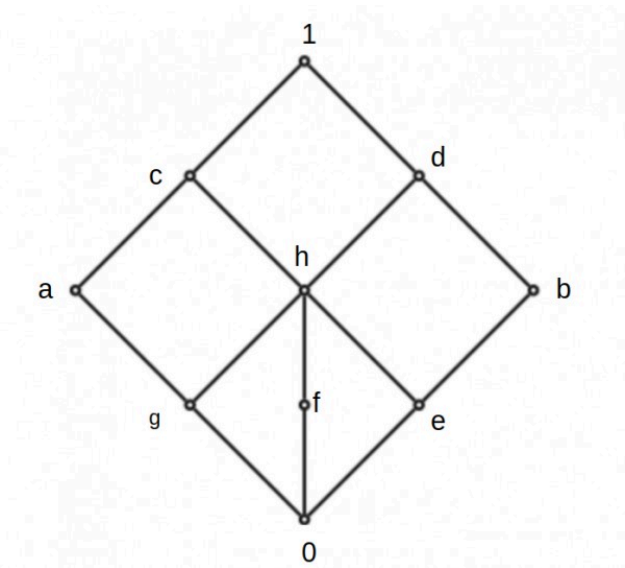
Q #63 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let $(L, \#)$ be a lattice, where $\#$ is some partial order relation defined on the base set L. Let $x, y, z \in L$. Which of the following is/are necessarily true?

A. $x \vee y = y \vee x$
B. $x \vee (x \wedge y) = x$
C. $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$
D. $x \wedge (x \vee y) = x$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #64 | Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra



The number of elements which do not have complements in the above lattices is _____

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #65 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let A, B be two predicate logic formulas. We say that "$A \models B$" (which means that "A infers B") iff for all interpretations in which A is true, B is also true.
So, "$A \models B$" iff "$A \rightarrow B$" is valid.

Which of the following inferences are valid in predicate logic?

A. $\forall x(P(x) \vee Q(x)) \models \forall x P(x) \vee \forall x Q(x)$
B. $\forall x(P(x) \rightarrow Q(x)) \models \forall x P(x) \rightarrow \forall x Q(x)$
C. $\forall x P(x) \rightarrow \forall x Q(x) \models \forall x(P(x) \rightarrow Q(x))$
D. $\neg \forall x(P(x) \wedge Q(x)) \models \exists x \neg P(x) \wedge \exists x \neg Q(x)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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